Debugging with Renode

Nucleo-144 STM32F767ZI · VS Code + Cortex-Debug · Renode 1.17

You need two windows: a **terminal** running Renode (the emulated board), and **VS Code** running the debugger. VS Code connects to Renode on port 3333, just as it would connect to OpenOCD and the ST-LINK on the real board.

Step 0 - Start clean (only if something is stuck)

- In VS Code, stop any debug session: **Shift+F5**.
- In any terminal running Renode, type `quit` (or press **Ctrl+C**).
- Check that nothing is still using port 3333. If this prints nothing, you are clean:

```
lsof -nP -iTCP:3333 -sTCP:LISTEN
```

- If it prints a line, that process (Renode or OpenOCD) is still running. Close it, or end it with `kill <PID>`, using the PID column.

Step 1 - Build the firmware

In the VS Code terminal, from the project folder:

```
make
```

Fix any compile errors before going on. Renode loads `build/final.elf`.

Step 2 - Start Renode (Terminal 1)

```
/Applications/Renode.app/Contents/MacOS/renode --console \
    -e "i @tools/renode/nucleo_f767zi.resc"
```

- Wait until you see: `GDB server with all CPUs started on port :3333`
- Leave this terminal open. Its `(nucleo_f767zi)` prompt is where you control the board.
- The CPU is **paused**. It hasn't run any code yet and waits for the debugger.

Do not use `make renode` for debugging. It also starts the CPU, so the code runs before you can stop it.

Step 3 - Set breakpoints in VS Code

Click in the left margin to add a red dot:

- on `button_init();` in `main()` (`app/src/main.c`)
- inside `EXTI15_10_IRQHandler` (`drivers/src/stm32f7xx_it.c`)

Step 4 - Attach the debugger

- Open **Run and Debug** (Cmd+Shift+D).
- Choose **Attach to Renode** from the dropdown and press **F5**.
- The debug toolbar appears. The CPU is still paused.
- Press **F5** again (Continue). The program runs and **stops at your breakpoint in main()**.
- Use **F10** (step over) and **F11** (step into) to walk through `button_init()`, then press **F5** to let it run.

Step 5 - Press the user button

Easiest: type this in the **Renode terminal** (Terminal 1). You don't need to pause VS Code:

```
sysbus.gpioPortC.UserButton1 PressAndRelease
```

VS Code should stop at the breakpoint in `EXTI15_10_IRQHandler`.

Alternative: from the VS Code **Debug Console** (Cmd+Shift+Y): press **F6** to pause, enter the line below, then press **F5**. `monitor` tells GDB to pass the command on to Renode.

```
monitor sysbus.gpioPortC.UserButton1 PressAndRelease
```

Each command presses the button **once**. Type it again for another press.

Step 6 - Look at registers and variables

Hover over variables, use the **Variables / Watch** panels, or type these in the Debug Console while paused:

```
p/x EXTI->PR            # bit 13 (0x2000) = EXTI13 pending
p/x EXTI->IMR           # bit 13 must be set
p/x EXTI->FTSR          # bit 13 = falling edge enabled
p/x SYSCFG->EXTICR[3]   # 0x20 = EXTI13 routed to port C
p/x GPIOC->MODER        # bits 27:26 must be 00 (input)
p/x GPIOC->IDR          # bit 13 = current button level
p/x NVIC->ISER[1]       # bit 8 = IRQ 40 (EXTI15_10) enabled
```

Step 7 - Stop

- VS Code: **Shift+F5** (or the red square).
- Renode terminal: type `quit`.

After you change the code

Simplest loop: **Shift+F5** in VS Code, `quit` in Renode, `make`, then repeat from Step 2. To reload without restarting Renode, type these in the Renode terminal and re-attach:

```
sysbus LoadELF @build/final.elf
cpu Reset
```

Debugger keys

Key	Action
F5	Start / Continue
F6	Pause
F10	Step over (run the whole line)
F11	Step into (enter the function)
Shift+F11	Step out (finish the current function)
Shift+F5	Stop debugging
Cmd+Shift+Y	Open the Debug Console

If something goes wrong

Problem	What to do
Attach fails: *could not connect* or timeout	Renode isn't running or hasn't finished loading. Start it (Step 2) and wait for the *port :3333* line.

Problem	What to do
Renode prints an error about `CreateServerSocketTerminal` or a port in use	Another Renode or OpenOCD is already running. Go to Step 0.
Attach to Renode missing from the dropdown	Check that `.vscode/launch.json` has that config and is saved.
The program runs before you can stop it	You started Renode with `make renode`. Use the command in Step 2 instead.
Debug Console says the target is running	Press **F6** to pause first, or type the command in the Renode terminal instead.
Handler breakpoint never hits	Check the Step 6 registers: IMR, FTSR, EXTICR[3], MODER, NVIC ISER[1]. Look for *WARNING* lines in the Renode terminal.
Handler runs again and again	The pending bit isn't cleared. The handler must write `EXTI->PR = EXTI_PR_PR13;`
Code changes don't show up	Run `make`, then restart or reload Renode.

Button edge: on the Nucleo-144, B1 is active-high, so pressing it gives a **rising** edge and releasing it a **falling** edge. With FTSR, the interrupt fires on **release**. `PressAndRelease` does both, so it always triggers. Use `Press` and `Release` separately to see each edge.